

Where does the neural DNO beat classical truncation?

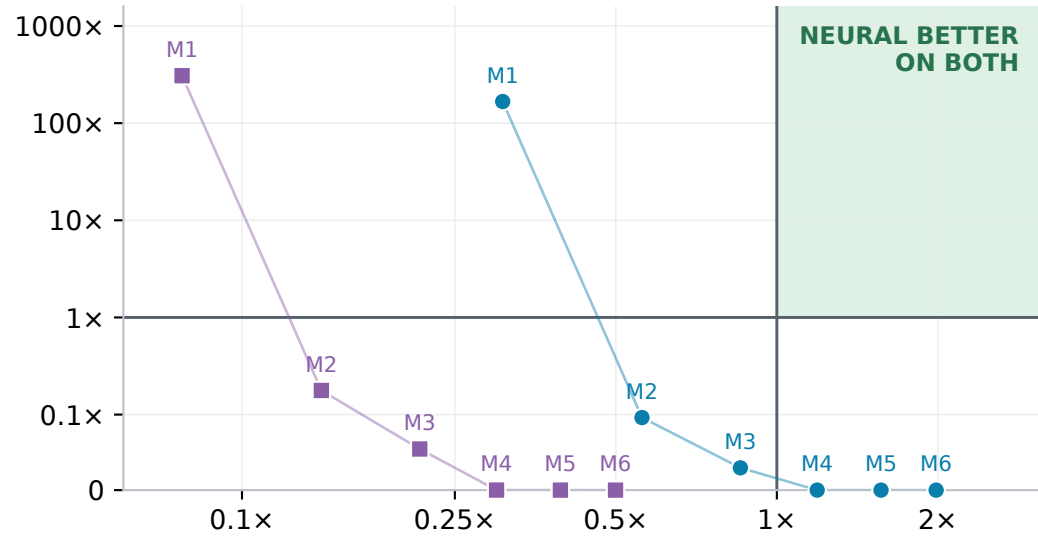
Right of 1x = lower recorded runtime. Above 1x = lower median surface error.

● N16: 16 branches/group (new) ■ N320: 320 branches/group

PROVISIONAL TIMING: classical excludes compilation/transfers; neural includes them. Not a matched speed benchmark.

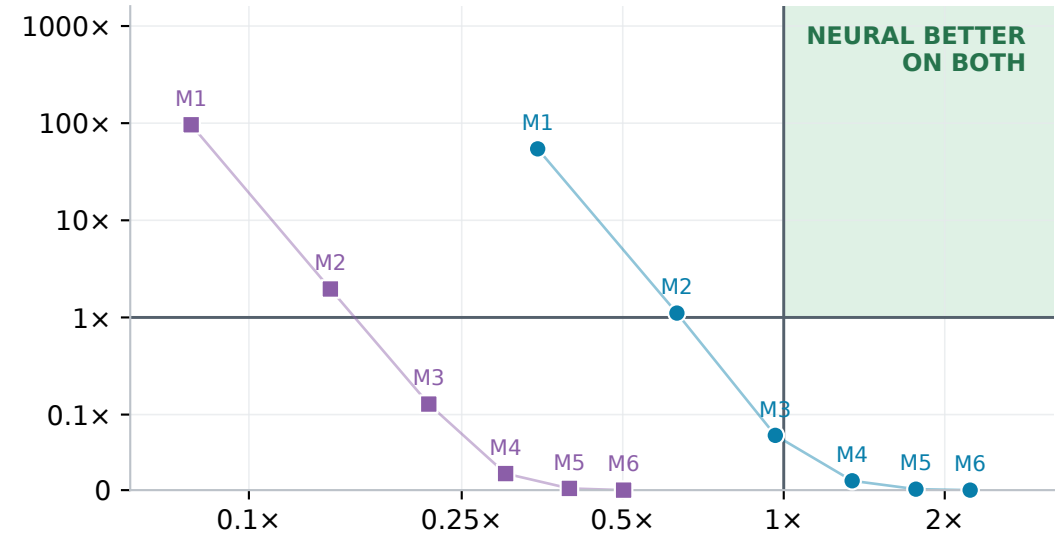
Stokes · T = 20

N320: 106.6 s / 0.00265% N16: 26.8 s / 0.00489%



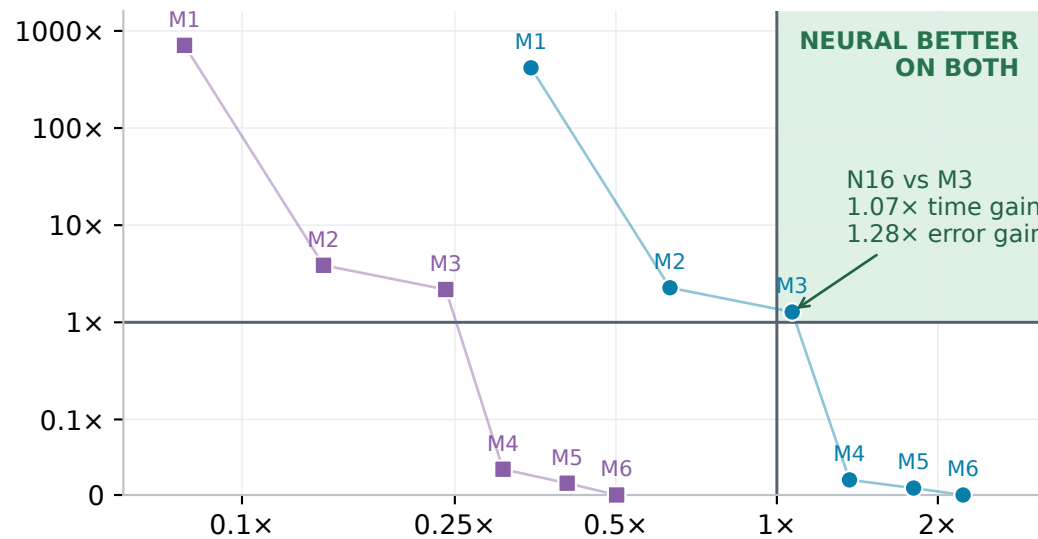
Tanaka · T = 200

N320: 1050.8 s / 0.0964% N16: 236.4 s / 0.171%



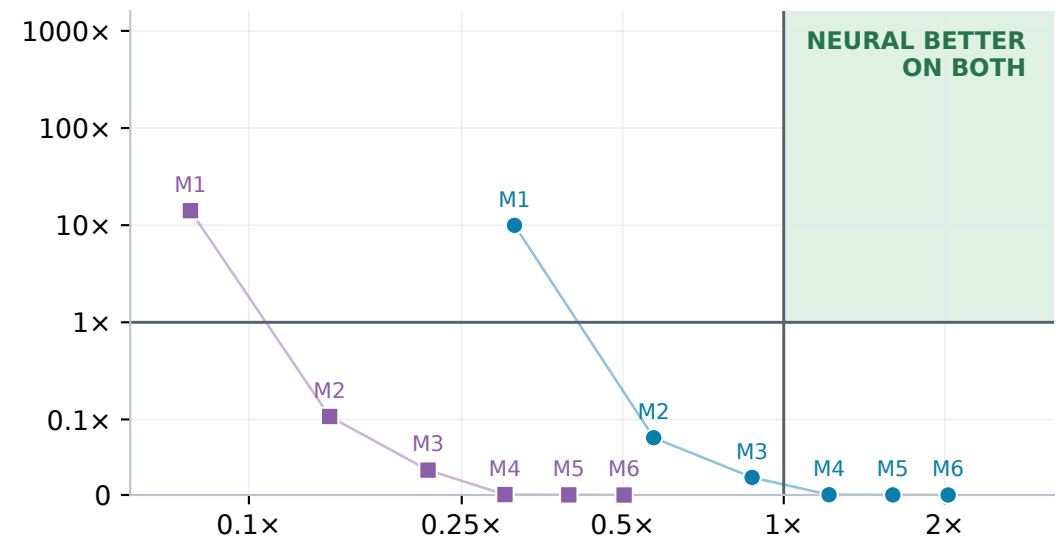
Benjamin-Feir · T = 200

N320: 1028.7 s / 0.104% N16: 231.5 s / 0.177%



JONSWAP / TMA · T = 20

N320: 105.9 s / 0.179% N16: 26.2 s / 0.253%



Same 32 held-out cases / family · batch-32 GPU rollout · terminal surface relative L^2 error vs M6 · $N = 1024$ · $dt = 0.01$

M6 has zero reference self-error, not zero exact-solution error. X is logarithmic; Y is linear below 0.1x and logarithmic above.

Panel subtitles give each neural model's recorded seconds / median error (%). Lines only connect tested orders; they are not fitted curves.